Virtual Stub Column Compression Test

Formulas used, where each one comes from, and where and how it produces the output · app page **Stub Column Test** (files `stub.py`, `stub_page.py`) · repository vectorverse11/model_spm

Source key

Key	Meaning
A	Your procedure / Excel sheet (EN 15512 Annex A.2.1)
B	Eurocode clause added by Claude (EN 1993-1-1 / 1-3 / 1-5)
C	Thin-walled section theory (textbook, no code clause)
D	Modelling choice by Claude (no standard prescribes it)
E	Alternative 3: finite strip method (CUFSM, Schafer & Ádány) + Direct Strength Method (AISI S100)

Clause numbers for EN 1993 were written from knowledge, not copied from the documents you supplied — please check them against your copies. Fixed constants: E = 210 000 N/mm², G = E / (2(1 + ν)) = 80 769 N/mm², ν = 0.3. Units: N, mm (loads shown in kN).

1 · User inputs on the Streamlit page

Group	Inputs	Used in
Specimen	W (web width), D (flange depth), t, lip length (0 = none), c (cold-formed length)	All section formulas, checks, buckling, shortening
Perforations	pitch p; holes across the web in one row n_w, width d_w; holes across each flange n_f, width d_f	5-hole check, A_{net}, effective strips
Cap / base plates	length, width, thickness of each	Projection check, cap thickness check, b
Material	f_y (typed), γ_{M0}; IS 2062 grade dropdown is reference only	ε, λ_p, ρ, A_{eff}, N_u, $\bar{\lambda}$, $A_{eff} = R_k/f_y$, $N_{b,Rd}$
Test	Alternative 1 or 2; dial least count LC (mm/div); load increment ΔN (kN)	Buckling lengths, load position, dial readings, load steps

2 · Formulas, sources, and how they produce the output

#	Formula	Source	Where in code	How it is used / what it produces
1	$c \geq 3 \cdot b_{max}$ (b_{max} = greatest flat width)	A — sheet item 1 / A.2.1.2 a)	specimen_checks()	Specimen-checks table ✓/✗ (warning only)
2	$c / p \geq 5$	A — sheet item 2 / A.2.1.2 a)	specimen_checks()	Specimen-checks table
3	$50 \leq p \leq 70$ mm	A — sheet item 2	specimen_checks()	Specimen-checks table
4	Cap Depth = Base Depth = $W + 2 \cdot t_{base}$ (calculated); (Depth − W)/2 ≥ 10 (along the web only)	A — sheet item 4 (per-side reading and web-only by user decision)	specimen_checks()	Specimen-checks table
5	$t_{cap} \geq 30$ mm	A — sheet item 5	specimen_checks()	Specimen-checks table
6	$b = c + 2 \cdot (t_{cap} + t_{base})$	A — sheet value b = 380 (formula is Claude's reading of it)	buckling_length_b()	Buckling length shown under the plate inputs; L_F for Alternative 1
7	$b_w = W - t$; $b_f = D - t$ (D − t/2 without lip); $b_l = d_{lip} - t/2$	C — centreline model	widths()	Flat widths for every later formula
8	$A_g = t \cdot \Sigma b_i$; $x_g = \Sigma(A_i x_i)/A_g$	C	gross_properties()	Step-by-step §2; load position (Alt. 2); Q
9	I_y, $I_z = \Sigma \int r^2\, dA$; $I_t = \Sigma\, b\, t^3/3$	C	gross_properties()	N_{cr} (formula 18–20)
10	ω(s) sectorial coordinate → shear centre x_0, $I_w = \int \omega^2\, dA$, $i_0^2 = (I_y+I_z)/A + x_0^2$	C — Vlasov thin-walled theory	gross_properties()	Torsional / torsional-flexural N_{cr}
11	$A_{net} = A_g - (n_w d_w + 2 n_f d_f) \cdot t$; x_{net}	A — 'minimum cross-section' named in procedure; formula C	net_area()	Step §2; Alt. 1 load range; proof load; start value of A_{eff} iteration
12	$\varepsilon = \sqrt{(235/f_y)}$; $\lambda_p = (b/t) / (28.4 \cdot \varepsilon \cdot \sqrt{k_\sigma}) \cdot \sqrt{(\sigma/f_y)}$	B — EN 1993-1-5 §4.4(2); σ-scaling: §4.4(4) / Annex E	rho_of()	Plate slenderness at each load step and at f_y
13	$k_\sigma = 4$ (internal: web, flange with lip); $k_\sigma = 0.43$ (outstand: lip, flange without lip)	B — EN 1993-1-5 Tables 4.1 / 4.2 (ψ = 1)	elements()	Input to λ_p
14	internal: $\rho = 1$ if $\lambda_p \leq 0.673$, else $\rho = (\lambda_p - 0.22)/\lambda_p^2$	B — EN 1993-1-5 §4.4(2) eq. 4.2 (ψ = 1)	rho_of()	Effective width of web and flanges
15	outstand: $\rho = 1$ if $\lambda_p \leq 0.748$, else $\rho = (\lambda_p - 0.188)/\lambda_p^2$	B — EN 1993-1-5 §4.4(2) eq. 4.3	rho_of()	Effective width of lips
16	internal: strips 0.5ρb at each edge; outstand: ρb from supported edge; central hole removes only its overlap with the strips → b_{eff}	B (strip layout, EN 1993-1-5 Table 4.1) + D (hole-overlap rule)	_strips(), effective_section()	Effective-width table; A_{eff}
17	$A_{eff} = t \cdot \Sigma b_{eff}$; x_{eff}; I_{eff}; $W_{eff} = I_{eff}/c_{max}$; $Q = A_{eff}/A_g$	B — EN 1993-1-3 §5.5.2 / §6.1.3; Q: EN 15512	effective_section()	A_{eff}, Q, W_{eff} shown in §3; used in N_u
18	Alt. 1: $L_F = b$, $L_T = c/2$; Alt. 2: $L_F = L_T = c$ (full length)	A — your sheet (EN 15512 A.2.1.4); Alt. 2 c/2 rule removed by user decision	critical_loads()	Buckling lengths in §4
19	$N_{cr,y} = \pi^2 E I_y / L_F^2$; $N_{cr,z} = \pi^2 E I_z / L_F^2$; $N_{cr,T} = (G \cdot I_t + \pi^2 E \cdot I_w / L_T^2)/i_0^2$	B — EN 1993-1-3 §6.2.3	critical_loads()	Critical loads in §4
20	$i_0^2(N-N_y)(N-N_z)(N-N_T) - N^2 x_0^2 (N-N_z) = 0$ → smallest root $N_{cr,TF}$; $N_{cr} = \min(\text{all})$	B — EN 1993-1-3 §6.2.3(7) (general cubic form)	critical_loads()	N_{cr} for $\bar{\lambda}$
21	$\bar{\lambda} = \sqrt{(A_{eff} \cdot f_y / N_{cr})}$	B — EN 1993-1-1 §6.3.1.2	failure_load()	Slenderness shown in §4
22	$\chi = 1$ if $\bar{\lambda} \leq 0.2$; else $\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$, $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)})$, $\alpha = 0.34$	B — EN 1993-1-1 §6.3.1.2(1),(4); curve b: EN 1993-1-3 Table 6.3; 0.2 limit also in your sheet	chi_of()	Reduction for buckling of the stub
23	$N_c = \chi \cdot A_{eff} \cdot f_y$	B — EN 1993-1-3 §6.1.3 / §6.2.1	failure_load()	Concentric failure load
24	Alt. 2: $e = 0$ → $N_u = N_c$	A (clamped platen, gross centroid central) + D ($e = 0$)	failure_load()	**Failure load, Alternative 2**

#	Formula	Source	Where in code	How it is used / what it produces
25	Alt. 1: $x_{load} = x_g + f(x_{net} - x_g)$, f = 0…1; e = $x_{load} - x_{eff}$; $N_u = \max_f 1/(1/N_c + \lvert e\rvert/(W_{eff} f_y))$	A (position range, 'adjusted to give max failure load') + B (EN 1993-1-3 §6.1.9 interaction) + D (sweep)	failure_load()	**Failure load, Alternative 1**
26	$\sigma = \min(N/A_{eff}(\sigma), f_y)$ iterated to convergence	D (with formulas 12–17)	_A_eff_at_load()	A_{eff} at every load step
27	Loads: $N_i = i \cdot \Delta N$, i = 0,1,… until $N \geq N_u$; last row at N_u	A (load in increments to failure) + D	run()	Rows of the readings table
28	$\delta = N \cdot c / (E \cdot A_{eff}(\sigma))$; Dial = δ / LC	D — elastic shortening	run()	Shortening (mm) and Dial (div) columns; both graphs
29	$\sigma_{lim,i} = f_y(\lambda_{lim}/\lambda_{p,i}(f_y))^2$, λ_{lim} = 0.673 / 0.748; $N_{proof} = \min \sigma_{lim} \cdot A_{net}$	D (definition) using B limits	proof_stress(), run()	Proof load metric and line on slope graph
30	(removed from the page) Slope1 = $N_{proof}/Dial_{proof}$; Slope2 = $(N_u - N_{proof})/(Dial_u - Dial_{proof})$; Slope3 = $N_u/Dial_u$	D — layout modelled on the vLab reference	run()	Slope lines and values (kN/div)
31	$R_k = N_u$	A — 'failure load' / 'maximum load recorded'	run()	**Failure load R_k metric**
32	$A_{eff} = R_k/f_y$; if $\bar{\lambda} > 0.2$ solve $\chi(A_{eff}) \cdot A_{eff} \cdot f_y = R_k$	A — your sheet ('area adjusted if slenderness > 0.2'); equation form B / EN 15512 A.2.1.4	a_eff_from_test()	**$A_{eff} = R_k/f_y$ metric**
33	$N_{b,Rd} = R_k / \gamma_{M0}$	A — your sheet	run()	**$N_{b,Rd}$ metric**
34	Strip shape functions: u, w-bending linear / Hermite across, sin(πy/a) along; K (membrane + plate bending), K_g (unit stress); K φ = λ K_g φ	E — CUFSM formulation (Schafer & Ádány 2006)	fsm.py	$\sigma_{cr}(a)$ for each half-wavelength a
35	Signature curve $\sigma_{cr}(a)$; 1st minimum → σ_{crl}, next → σ_{crd}; $P_{crl} = \sigma_{crl} A_g$, $P_{crd} = \sigma_{crd} A_g$	E — CUFSM method	dsm_analysis()	Signature-curve graph; P_{crl}, P_{crd}
36	$\lambda_c = \sqrt{(P_y/P_{cre})}$; $P_{ne} = 0.658^{\lambda_c^2} P_y$ ($\lambda_c \leq 1.5$) or $0.877 P_y/\lambda_c^2$; $P_{ne} \leq P_{ynet} = A_{net} f_y$	E — AISI S100 DSM (global); P_{cre} from formulas 19–20 with L = c	dsm_analysis()	P_{ne}
37	$\lambda_l = \sqrt{(P_{ne}/P_{crl})}$; $P_{nl} = P_{ne}$ ($\lambda_l \leq 0.776$) else $[1 - 0.15(P_{crl}/P_{ne})^{0.4}](P_{crl}/P_{ne})^{0.4} P_{ne}$; ≤ P_{ynet}	E — AISI S100 DSM (local)	dsm_analysis()	P_{nl}
38	$\lambda_d = \sqrt{(P_y/P_{crd})}$; P_{nd} with net-section limits $\lambda_{d1} = 0.561 P_{ynet}/P_y$, $\lambda_{d2} = 0.561(14(P_y/P_{ynet})^{0.4} - 13)$, else $[1 - 0.25(P_{crd}/P_y)^{0.6}](P_{crd}/P_y)^{0.6} P_y$	E — AISI S100 DSM (distortional, members with holes)	dsm_analysis()	P_{nd} (used only if ticked)
39	$N_u = P_n = P_{nl}$ (or $\min(P_{nl}, P_{nd})$ if distortional ticked)	E + D (distortional optional: EN 15512 test excludes it)	failure_load()	**Failure load, Alternative 3**
40	$A_{eff}(\sigma) = A_g \cdot [1 - 0.15(\sigma_{crl}/\sigma)^{0.4}](\sigma_{crl}/\sigma)^{0.4} \leq A_{net}$; $N_{proof} = 0.776^2 \sigma_{crl} A_{net}$	D — DSM local curve used as an effective-area ratio	_A_eff_at_load(), run()	Alt. 3 load–dial curve and proof load
41	$y_{CG} = \Sigma(A_i x_i)/\Sigma A_i = b_f(b_f + 2b_l)/(b_w + 2b_f + 2b_l)$ (or user value)	C — statics / EN 1993-1-3 Annex C; custom option by user decision	failure_load()	CG on the section diagram; load position (Alt. 1 range start, Alt. 2/3)
42	f_y = ReH of the IS 2062 grade for the thickness band of t (or typed if Customize)	IS 2062:2011 Table 2	catalog.grade_yield()	Yield stress used everywhere
43	$w = e \cdot [\sec(\pi/2 \cdot \sqrt{(N/N_{cr})}) - 1]$; failure: $N/N_c + N \cdot e \cdot \sec(\pi/2 \cdot \sqrt{(N/N_{cr})})/(W_{eff} f_y) = 1$	B — secant formula (classical) with EN 1993-1-3 §6.1.9-type interaction; D (e from user)	eccentric_test()	Eccentric Load vs Displacement graph and eccentric failure load

3 · How the formulas chain into the output

Stage	Formulas	Output on the page
Inputs → checks	1–6	Specimen-checks table (✓/✗), buckling length b
Section	7–11	Step-by-step §2: widths, A_g, A_{net}, I, I_t, I_w, shear centre
Local buckling	12–17 (at σ = f_y)	Effective-width table, A_{eff}, x_{eff}, W_{eff}, Q (§3)
Stub buckling	18–23	§4: L_F, L_T, N_{cr}, $\bar{\lambda}$, χ, N_c
Failure	24 or 25 (Alt. 3: 34–39)	§5: failure load N_u (green banner); Alt. 3 adds the signature curve (§3)
Virtual loading	26–28 (12–17 at every step)	Readings table, Load vs Dial graph, CSV
Curve features	29–30	Proof load, Slope1–3 graph
EN 15512 results	31–33	Metrics R_k, A_{eff}, $N_{b,Rd}$; results CSV

4 · Worked example (numbers produced by the app engine)

W 110, D 90, t 2, lip 20, c 300, p 60, 2 × 10 mm web holes, 1 × 8 mm hole per flange, cap/base 130 × 100, t_{cap} 30, t_{base} 10, f_y 350, γ_{M0} 1.0, LC 0.01 mm/div, ΔN 1 kN. Lip and hole sizes are placeholders.

Quantity	Formula #	Alternative 1	Alternative 2
b	6	380 mm	380 mm
Flat widths b_w / b_f / b_l	7	108 / 88 / 19 mm	same
A_g / A_{net}	8, 11	644.0 / 572.0 mm²	same
λ_p web / flange / lip (at f_y)	12	1.160 / 0.945 / 0.623	same
ρ web / flange / lip	14, 15	0.698 / 0.812 / 1.000	same
A_{eff} (at f_y), Q	17	512.56 mm², 0.796	same
L_F / L_T	18	380 / 150 mm	300 / 300 mm
N_{cr}	19, 20	10,529 kN	4,488 kN
$\bar{\lambda}$ → χ	21, 22	0.131 → 1.000	0.200 → 1.000
$N_c = \chi A_{eff} f_y$	23	179.40 kN	179.40 kN
e	24 / 25	-1.265 mm	0
Failure load $N_u = R_k$	24 / 25, 31	**170.72 kN**	**179.40 kN**
Shortening / dial at failure	28	0.4656 mm / 46.6 div	0.5000 mm / 50.0 div
Proof load	29	67.36 kN	67.36 kN
Slope1 / 2 / 3 (kN/div)	30	3.966 / 3.495 / 3.667	3.966 / 3.393 / 3.588
$A_{eff} = R_k/f_y$	32	487.78 mm²	512.56 mm²
$N_{b,Rd} = R_k/\gamma_{M0}$	33	170.72 kN	179.40 kN

5 · Limitations and points to confirm

- Failure is calculated (formulas 23–25), then the loading loop stops at N_u; there is no falling branch after the peak — EN 15512 only records the maximum load.
- No initial imperfections or residual stresses are modelled, so the curve is straighter near failure than a physical test. Calibration against one real test is recommended.
- Flange-with-lip is treated as an internal plate (distortional buckling is excluded by the test purpose). Corner radii are ignored (square-corner centreline).
- Holes are taken in one cross-section (conservative). The hole-overlap rule (16) is a modelling choice.
- Alternative 2 is treated as concentric (e = 0). The 0.5 % holding load is described in the procedure but the app currently starts the readings at 0 kN.

- IS 2062 dropdown is reference only; the typed f_y drives every strength result. E, G, ν are fixed steel constants.
- EN 1993 clause numbers are quoted from knowledge — verify against your copies.